

Software Testing & Continuous Integration

Individual Laboratory Report - Phases 1, 2, and 3

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Course	CS 472 - Software Product Design
Team	Group 2
Submission date	September 18, 2026
Team repository	github.com/kenner1-unlv/Group-2
Assignment	Software Testing & Continuous Integration Lab

Completion summary

3 substantive contributions merged | 3 evidence-based peer reviews completed | 3 phases documented

Report Overview

This report documents my individual work across all three required phases: improving test coverage, developing an API feature with RED-GREEN-REFACTOR, and extending GitHub Actions with independently visible CI jobs. Each section includes local evidence, direct issue and pull-request links, implementation excerpts, peer-review evidence, and final merge references.

Repository and synchronization

The authoritative Group 2 repository is hosted under my GitHub account. GitHub does not permit an account to fork its own repository, so I completed my work through uniquely named protected feature branches and synchronized them against **origin/main** before integration. The evidence below shows a clean working tree, the configured team remote, and the final merged history.

```

russell@New_Huttness MINGW64 ~/repos/Group-2 (main)
$ git switch main
git pull --ff-only origin main
git status
git log -1 --oneline
Already on 'main'
Your branch is up to date with 'origin/main'.
remote: Enumerating objects: 288, done.
remote: Counting objects: 100% (126/126), done.
remote: Compressing objects: 100% (74/74), done.
remote: Total 288 (delta 76), reused 63 (delta 46), pack-reused 162 (from 1)
Receiving objects: 100% (288/288), 49.77 KiB | 621.00 KiB/s, done.
Resolving deltas: 100% (152/152), completed with 2 local objects.
From https://github.com/kenned1-unlv/Group-2
* branch      main      -> FETCH_HEAD
29aa705..68e6578 main    -> origin/main
Updating 29aa705..68e6578
Fast-forward
 .github/workflows/ci.yml      | 100 ++++++
 README.md                    | 22 +
 ci_lab/tests/test_counter.py | 5 ++
 tdd_lab/src/counter.py       | 30 ++++++
 tdd_lab/src/status.py        | 1 +
 tdd_lab/tests/test_counter.py | 52 ++++++
 test_coverage_lab/tests/test_account.py | 127 ++++++
 7 files changed, 320 insertions(+), 17 deletions(-)
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
68e6578 (HEAD -> main, origin/main, origin/HEAD) Merge pull request #57 from kenned1-unlv/ci-split-jobs-kenned1-unlv
russell@New_Huttness MINGW64 ~/repos/Group-2 (main)
$

```

Figure 1. Fast-forward synchronization to the final main branch; the working tree is clean and current with origin/main.

Repository Evidence

```

MINGW64~/c:/Users/russe/repos/Group-2
AuthorDate: Mon Sep 14 18:27:11 2026 -0700
Commit:      kenner1-unlv <kenner1@unlv.nevada.edu>
CommitDate: Mon Sep 14 18:22:13 2026 -0700

REFACTOR: extract counter reset helper

diff --git a/tdd_lab/src/counter.py b/tdd_lab/src/counter.py
index a8006bc..3561852 100644
--- a/tdd_lab/src/counter.py
+++ b/tdd_lab/src/counter.py
@@ -12,8 +12,11 @@ def counter_exists(name):
     """Check if counter exists"""
     return name in COUNTERS

-def reset_counters():
-    """Reset every stored counter to zero"""
-    for name in COUNTERS:
-        COUNTERS[name] = 0
+def reset_counters():
+    """Reset every stored counter to zero"""
+    for name in COUNTERS:
+        COUNTERS[name] = 0

@app.route('/counters/<name>', methods=['POST'])
def create_counter(name):
    """Create a counter"""
@@ -28,6 +33,5 @@ def list_counters():
@app.route('/counters/reset', methods=['POST'])
def reset_all_counters():
    """Reset all counters to zero"""
    for name in COUNTERS:
        COUNTERS[name] = 0
    reset_counters()
    return jsonify({"message": "All counters reset"}), status.HTTP_200_OK

russe@New_Holiness MINGW64 ~/repos/Group-2 (main)
$ cd /c:/Users/russe/repos/Group-2
git remote -v
git branch --show-current
git status
git log --oneline --decorate -8
origin https://github.com/kenner1-unlv/Group-2.git (fetch)
origin https://github.com/kenner1-unlv/Group-2.git (push)
main
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
88e6578 (HEAD -> main, origin/main, origin/HEAD) Merge pull request #57 from kenner1-unlv/ci-split-jobs-kenner1-unlv
292e376 Merge main and integrate all CI enhancements into split jobs
0060751 Merge pull request #82 from ethanvfour/phase3-dependency-caching-ethanvfour
771e5b3 Merge pull request #80 from ethanvfour/phase2-invalid-counter-name-ethanvfour
a839416 Merge pull request #78 from ethanvfour/phase1-positive-deposit-ethanvfour
ca1ad22 Merge pull request #65 from jincnaz29/p1-withdraw-insuff-funds
d604826 Merge pull request #67 from geeklay/ci_label
a7832c1 Add docstring to withdraw test and synced with main

russe@New_Holiness MINGW64 ~/repos/Group-2 (main)
$

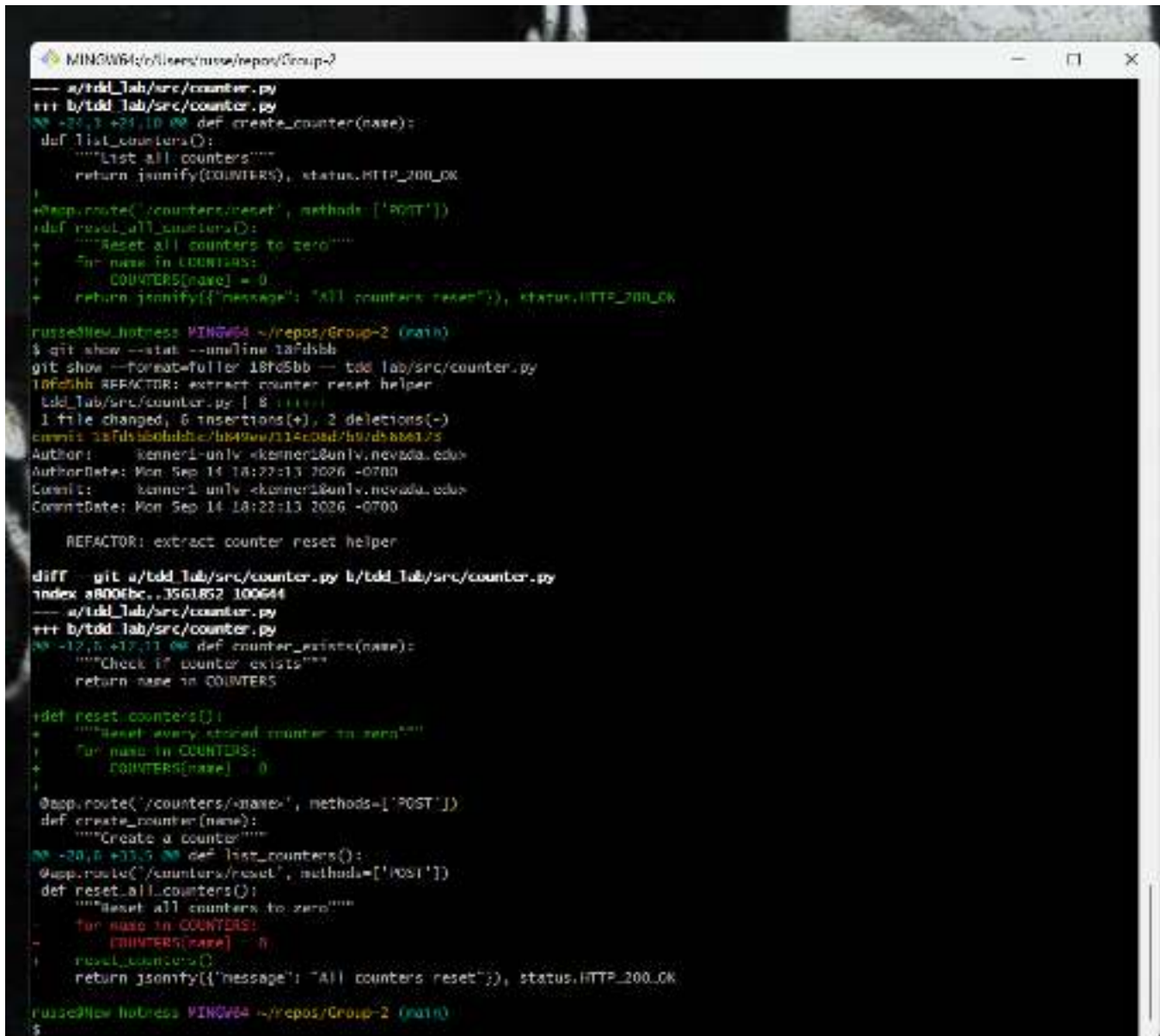
```

Figure 2. Origin remote, clean main branch, and recent merge history including my final Phase 3 merge.

Branch-protection workflow

Issue -> uniquely named branch -> focused commits -> pull request -> assigned technical review -> CI checks -> author merge

Phase 2 - RED



```
MINGW64~/repos/russ/repo/Group-2
--- a/tdd_lab/src/counter.py
+++ b/tdd_lab/src/counter.py
@@ -20,1 +21,10 @@ def create_counter(name):
+ def list_counters():
+     """List all counters"""
+     return jsonify(COUNTERS), status.HTTP_200_OK
+
+ @app.route('/counters/reset', methods=['POST'])
+ def reset_all_counters():
+     """Reset all counters to zero"""
+     for name in COUNTERS:
+         COUNTERS[name] = 0
+     return jsonify({'message': 'All counters reset'}), status.HTTP_200_OK

russ@New hotness MINGW64 ~/repos/Group-2 (main)
$ git show --stat --unified 18f5bb
git show --format=fuller 18f5bb -- tdd_lab/src/counter.py
18f5bb REFACTOR: extract counter reset helper
tdd_lab/src/counter.py | 8 +++++
1 file changed, 6 insertions(+), 2 deletions(-)
commit 18f5bb0b5d3e/b549ee/114e0ad/b9/d5886173
Author: kenneri-unlv <kenneri@unlv.nevada.edu>
AuthorDate: Mon Sep 14 18:22:13 2026 -0700
Commit: kenneri-unlv <kenneri@unlv.nevada.edu>
CommitDate: Mon Sep 14 18:22:13 2026 -0700

    REFACTOR: extract counter reset helper

diff --git a/tdd_lab/src/counter.py b/tdd_lab/src/counter.py
index a800fbc..3561852 100644
--- a/tdd_lab/src/counter.py
+++ b/tdd_lab/src/counter.py
@@ -12,5 +17,11 @@ def counter_exists(name):
     """Check if counter exists"""
     return name in COUNTERS

+ def reset_counters():
+     """Reset every stored counter to zero"""
+     for name in COUNTERS:
+         COUNTERS[name] = 0
+
+ @app.route('/counters/<name>', methods=['POST'])
+ def create_counter(name):
+     """Create a counter"""
@@ -20,6 +33,5 @@ def list_counters():
+ @app.route('/counters/reset', methods=['POST'])
+ def reset_all_counters():
+     """Reset all counters to zero"""
+     for name in COUNTERS:
+         COUNTERS[name] = 0
+     reset_counters()
+     return jsonify({'message': 'All counters reset'}), status.HTTP_200_OK

russ@New hotness MINGW64 ~/repos/Group-2 (main)
$
```

Figure 6. RED commit 8d98d99 adds the reset-all test before any endpoint implementation.

Recorded RED result

```
assert result.status_code == status.HTTP_200_OK E assert 201 == 200 1 failed
```

The failure was meaningful: without a dedicated reset endpoint, Flask matched the existing dynamic POST route and created a counter named reset, returning 201 instead of the required 200.

Phase 2 - Review and Outcome

Final verification recorded in PR #62

Targeted reset test: passed Full suite at contribution time: 3 passed Total coverage: 97% Behavior preserved after refactoring

Reviewer feedback received

The implementation later required synchronization because multiple Phase 2 branches modified the same shared Counter API files. I merged current upstream changes, preserved the reset behavior and tests, reran the suite, and obtained approval on the synchronized branch before merging. This reinforced that resolving integration conflicts and revalidating behavior are part of the TDD workflow, not separate administrative tasks.

Technical peer review

I reviewed Sokrat Rostomyan's [Phase 2 PR #75](#) and submitted an [evidence-based approval](#). I checked out RED commit fa5827b and reproduced the 405-versus-200 failure, then verified GREEN commit 60dc341, the complete 3-test suite, 96% coverage, focused scope, and the reasonable no-refactor justification.

Phase 3 - Continuous Integration

Contribution: split CI into independent jobs

Setup	Issue #51	PR #52
My work	Issue #53	PR #57
Merge	Merge commit 68e6578	

I replaced one combined CI job with separate **Lint**, **Test**, and **Coverage** jobs. This lets unaffected checks complete independently and makes the failing quality gate immediately visible.

Final integrated workflow excerpt

```
jobs: lint: flake8 src --count --select=E9,F63,F7,F82 ... test: pytest --no-cov # Python 3.9, 3.10, 3.11
matrix coverage: pytest --cov=src --cov-fail-under=80 --cov-report=json upload-artifact: coverage-report
```

```
[...]
$ cat > test_session_start
$ cd test_job
$ pytest --no-cov

platform linux -- python 3.10.7, pytest-6.2.5, pluggy-1.0.0 -- /usr/bin/python3.10
cachedir: .pytest_cache
rootdir: /home/runner/repos/group-2/ci_lab
configfile: pytest.ini
plugins: cov-3.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 91%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_prevent Updating non-existent counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter_if_exists PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter_by_value PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_reseting_non-existent counter PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshold PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold PASSED [ 100%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [ 100%]

===== 22 passed in 0.13s =====

[.]
$ cat > lint_job
$ flake8 src --count --select=E9,F63,F7,F82 --show-source --statistics
0

[.]
$ cat > merge_commit_68e6578
```

Figure 9. Local Windows evidence: 22/22 plain tests pass and Flake8 reports zero selected errors.

Phase 3 - Local Coverage and Baseline

```

PASSED [ 95%]
tests/test_counter.py::testCounterEndpoints::test_validate_counter_name PASSED [
100%]

----- tests coverage -----
coverage: platform win32, python 3.14.7-final-0
Name           Stats    Miss  Cover   Missing
-----
src\__init__.py    3      0   100%
src\counter.py   87      5    94%   71, 85, 96, 99-100
src\status.py     7      0   100%
TOTAL            97      5    95%
22 passed in 0.18s

(.venv)
russak@ms-holmes: PING64 ~/repos/Group-2/ci_lab (main)
$ deactivate
russak@ms-holmes: PING64 ~/repos/Group-2/ci_lab (main)
$

```

Figure 10. Local Windows coverage run: 22 tests pass with 95% total coverage.

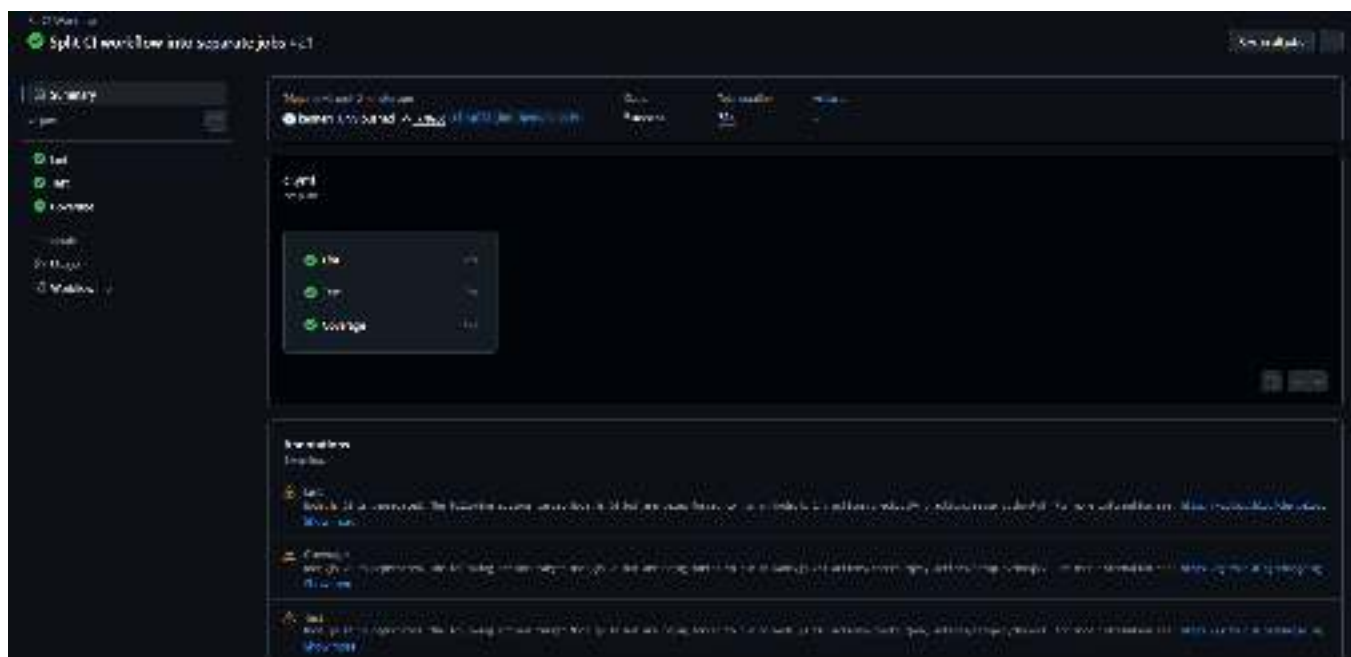


Figure 11. Successful baseline run #21 visibly separates Lint, Test, and Coverage.

Direct evidence: [GitHub Actions run #21](#).

Phase 3 - Controlled Failure

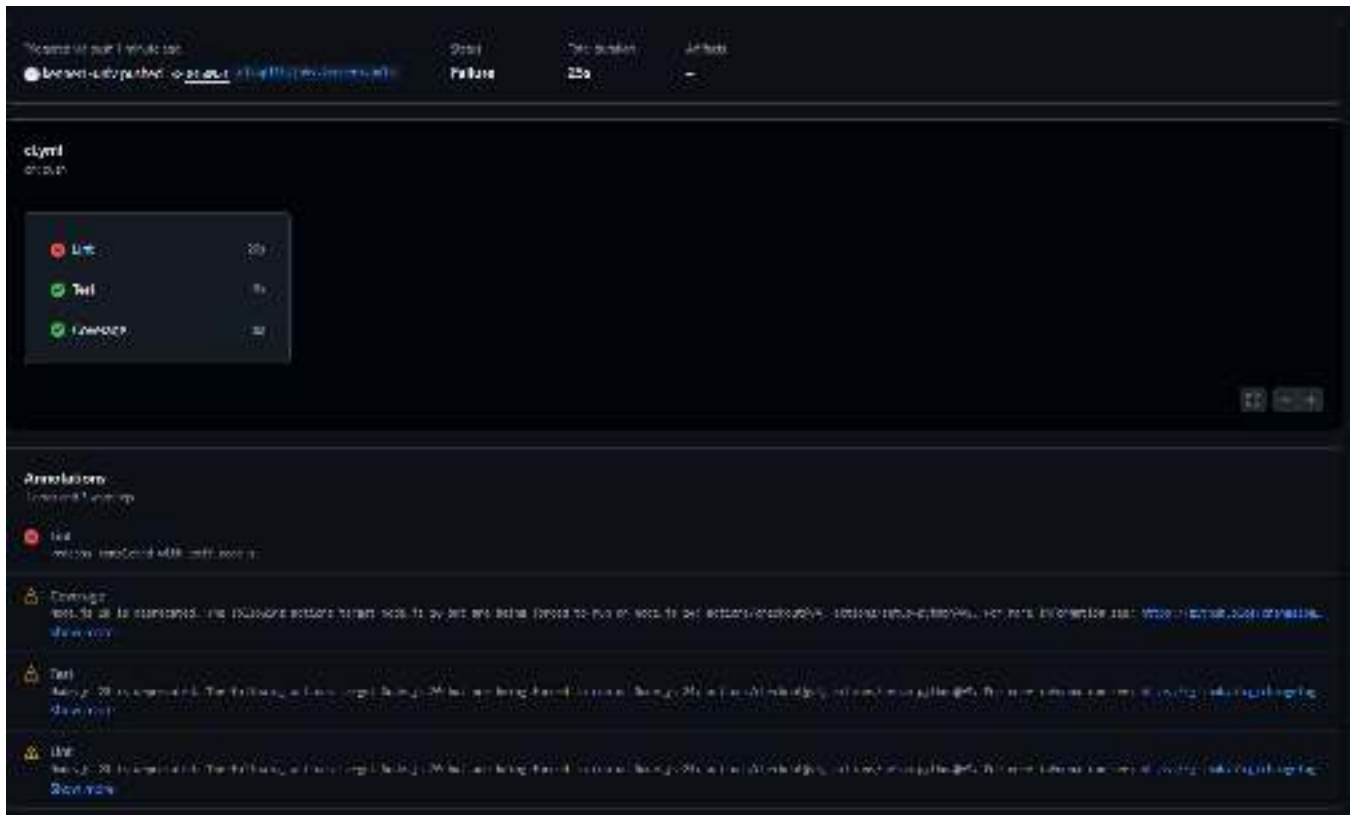


Figure 12. Controlled failure run #22: Lint is red while Test and Coverage remain green.

Direct evidence: [GitHub Actions run #22](#).



Figure 13. Flake8 log excerpt identifying nonexistent_directory and E902 FileNotFoundError.

Root-cause analysis

The controlled commit changed Flake8's valid target from src to nonexistent_directory. Flake8 could not traverse the path, raised E902 FileNotFoundError, and returned exit code 1. Because the jobs were independent, Test and Coverage still completed successfully.

Phase 3 - Fix, Recovery, and Integration

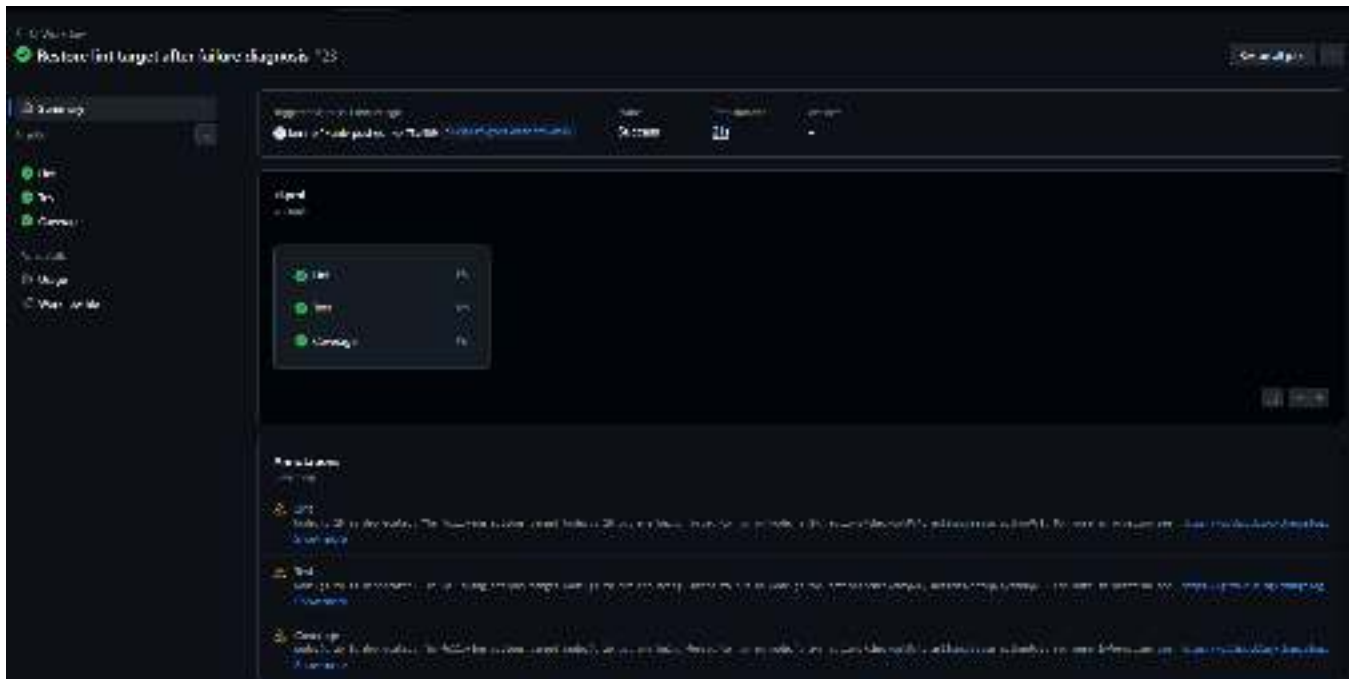


Figure 14. Recovery run #23 after restoring the Flake8 target to src; all three jobs are green.

Direct evidence: [GitHub Actions recovery run #23](#).

Before final merge, I synchronized the branch with current main and resolved repeated workflow conflicts. The final resolution preserved every merged team enhancement: the Python 3.9/3.10/3.11 matrix, dependency caching, the 80% coverage gate, JSON coverage artifact upload, README documentation, and my split-job design.

The final integrated [run #77](#) passed five visible jobs: Lint, Test 3.9, Test 3.10, Test 3.11, and Coverage. The artifact-upload step also passed.

Phase 3 - Peer Review

I reviewed Sokrat Rostomyan's [Phase 3 PR #85](#) and submitted an [evidence-based approval](#).

Evidence examined

- Workflow diff adding --cov-fail-under=80.
- Controlled 100% threshold run in which all 22 tests passed but 94.85% coverage correctly failed the gate.
- CI log proving the nonzero status came from the unmet coverage threshold rather than a failed test.
- Corrected 80% threshold and successful PR-triggered run #48.
- Focused one-line final diff and issue #70 acceptance criteria.

Reviewer feedback received

Astrid initially requested changes because upstream Phase 3 work created a workflow merge conflict. I synchronized twice as additional CI contributions merged, combined all enhancements instead of overwriting them, posted fresh CI evidence, and obtained a new approval on the exact final commit. PR #57 then merged successfully.

Phase 3 result
Issue #53 closed PR #57 approved and merged merge commit 68e6578 final integrated CI green

Reflection

This lab made the distinction between test quantity, coverage, and test quality concrete. Coverage tells me which statements executed, but it does not prove that the assertions are meaningful. In Phase 1, the useful evidence was not only that lines 94 and 98 became covered; it was that the test explicitly verified both state transitions. In Phase 2, preserving separate RED, GREEN, and REFACTOR commits created an auditable record that the test genuinely preceded the implementation and failed for the intended reason.

The CI phase showed why pipeline structure matters operationally. A single red workflow says that something failed; independent jobs say what failed and preserve evidence from unaffected checks. The controlled Flake8 failure demonstrated this directly: Lint failed while Test and Coverage remained green. Reading the E902 log turned a generic red status into a precise root cause and a focused correction.

The most important collaborative lesson was that technical review is part of system throughput. CI configuration and API files were shared integration points. Each delayed merge increased the chance that open branches would become stale, approvals would be dismissed, and contributors would repeat conflict resolution. A useful review therefore needs to be prompt as well as evidence-based. At the same time, the final conflict work was valuable: I had to preserve the intent of several independent enhancements rather than simply make my branch win.

Overall, I leave the lab with a more professional model of software quality: define observable acceptance criteria, produce focused evidence, preserve the history of the reasoning, review the exact integrated change, and treat a green pipeline as a reproducible claim rather than a decorative badge.

Individual completion

Phase 1 PR #61 merged + review of #72 Phase 2 PR #62 merged + review of #75 Phase 3 PR #57 merged + review of #85

Evidence Link Appendix

Requirement	Issue / PR	Final evidence
Phase 1 setup	Issue #43	PR #44
Phase 1 contribution	Issue #59	PR #61
Phase 1 review	Sokrat PR #72	Review comment
Phase 2 setup	Issue #46	PR #47
Phase 2 contribution	Issue #60	PR #62
Phase 2 review	Sokrat PR #75	Review comment
Phase 3 setup	Issue #51	PR #52
Phase 3 contribution	Issue #53	PR #57
Baseline		Actions run #21
Controlled failure		Actions run #22
Recovery		Actions run #23
Final integration		Actions run #77
Phase 3 review	Sokrat PR #85	Review comment

Submission statement

This single PDF covers all three phases and is intended to be committed to the Group 2 repository and uploaded unchanged to Canvas, as required by the assignment.